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AgriScope

A Pixel-Grounded Multimodal Model and 11-Million-Sample Dataset Bring Precision to Agricultural Image Understanding

By Abderrahmene Boudiaf, Mohamad Alanssari, Irfan Hussain & Sajid Javed (Khalifa University of Science and Technology)

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Agricultural AI gets its first unified pixel-grounded multimodal model. *AgriScope* jointly supports image-level classification, region-level referring segmentation, and pixel-level grounded caption generation within a single framework trained on AgriGround—a new 500K-image, 11-million-sample dataset—lifting grounded caption CIDEr from 107.9 to 118.6 and mIoU from 55.0 to 59.4 over the best prior baseline on the accompanying benchmark.

AT A GLANCE

Problem: Agricultural image understanding requires fine-grained pixel-level localisation (disease lesions, pest bodies, growth stage features) that existing text-only MLLMs cannot provide.

Why it matters: Actionable recommendations for precision agriculture (targeted spraying, selective harvesting) require knowing *where* in the image the problem is, not just *what* it is.

Method: Biological-semantic encoder pre-trained on botanical taxonomies; multi-scale dense spatial features (FPN over BSE and DINOv2); mask transformer pixel decoder; AgriGround instruction- tuning dataset with 500K images and 11M samples.

Result: GCG CIDEr 118.6 (+10.7), GCG mIoU 59.4 (+4.4), referring seg F1 73.2 (+5.4) over PixelLM baseline.

Evidence: ICML 2026 evaluation across 6 agricultural task types; ablations over encoder, decoder, and dataset scale.

The Big Picture

Precision agriculture aims to deliver the right intervention (pesticide, fertiliser, pruning) to exactly the right location at exactly the right growth stage. This requires a visual understanding system that can answer not just “does this plant have rust disease?” but “which leaves show early-stage rust and where on each leaf are the uredinia?”

Multimodal large language models have advanced rapidly but their outputs are purely textual: they name diseases

and pests but do not localise them in pixel coordinates. AgriScope closes this gap with a unified model that takes an agricultural image (or image sequence) plus a natural-language instruction and outputs both a text response *and* pixel-level segmentation masks for all referenced visual entities.

Why Existing Approaches Fall Short

Text-only MLLMs (GPT-4V, InternVL-2, LLaVA) produce written diagnoses but cannot drive autonomous machinery that requires pixel masks or bounding boxes. Their agricultural knowledge is also limited by training on general web data rather than domain-curated imagery.

Specialist agricultural classifiers (PlantVillage CNNs, domain-specific ViTs) excel at narrow tasks (identify this disease from a leaf patch) but cannot answer open-ended agricultural questions, handle novel species, or produce grounded multi-entity outputs.

Grounding MLLMs (LISA, PixelLM) support referring segmentation for general-domain images but use standard ViT features that lack biological specificity: they cannot distinguish powdery mildew from leaf wax deposits, or aphid clusters from normal trichomes.

Core Mechanism

Biological-Semantic Encoder (BSE). A ViT-L fine-tuned on 2.3M agricultural images annotated with hierarchical biological labels (kingdom, class, order, family, genus, species, disease type, disease stage). The BSE produces feature tokens $\{f_i^{\text{bio}}\}$ embedding biological category memberships alongside spatial features.

Dense Spatial Representations (DSR). A multi-scale feature pyramid fuses BSE tokens with a frozen DINOv2 encoder:

$$F_{\text{DSR}} = \text{FPN}(f^{\text{bio}}, f^{\text{dino}}) \in \mathbb{R}^{C \times H/4 \times W/4},$$

providing spatially dense features at three resolutions for both coarse recognition and fine lesion localisation.

Pixel Decoder. A mask transformer decoder takes query embeddings $\{q_k\}$ derived from the LLM output’s [SEG] tokens and predicts binary masks:

$$M_k = \text{PD}(q_k, F_{\text{DSR}}) \in \{0, 1\}^{H \times W}.$$

When the LLM generates [SEG], the last hidden state at that position feeds directly into the pixel decoder as q_k .

Technical Formulation

Training minimises:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{dice}} \mathcal{L}_{\text{dice}},$$

with $\lambda_{\text{BCE}} = 1.0$, $\lambda_{\text{dice}} = 0.5$.

Binary cross-entropy on predicted mask $\hat{M}_k$ vs. ground-truth M_k^* :

$$\mathcal{L}_{\text{BCE}} = - \sum_p [M_k^*(p) \log \hat{M}_k(p) + (1 - M_k^*(p)) \log (1 - \hat{M}_k(p))].$$

Dice loss for shape overlap:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_p M_k^*(p) \hat{M}_k(p)}{\sum_p M_k^*(p) + \sum_p \hat{M}_k(p)}.$$

Learning or Inference Procedure

1. Pre-train BSE on 2.3M agricultural images with hierarchical biological classification (warm-start from ViT-L ImageNet checkpoint).
2. Train AgriScope on AgriGround for 3 epochs with frozen DINOv2 and trainable BSE, LLM adapters (LoRA), and pixel decoder.
3. Optional task-specific fine-tuning for 1 epoch on narrow subsets (e.g., disease detection only).
4. Inference: input image + instruction; LLM generates text interleaved with [SEG] tokens; pixel decoder predicts a mask for each [SEG].

AgriGround construction: 500K source images from PlantVillage, iNaturalist-Agriculture, FGVC-Plant, and CropDiseaseDB; captions generated by GPT-4V with botanical system prompts; grounded with Grounding-DINO + SAM2; 11M instruction-following samples across 6 task types (disease, pest, species, growth, weed, nutritional deficiency).

What the Guarantee Says

No formal guarantees. The ICML reviewers cited the breadth of the AgriGround evaluation: 6 task types, 3 cross-dataset generalisation settings, and both in-distribution and zero-shot conditions, providing strong empirical evidence that improvements are not evaluation artefacts.

Experimental Findings

Main benchmark: AgriGround-Test (10K held-out samples).

Method	CIDEr	mIoU	Seg F1	VQA Acc
LLaVA-1.5	82.3	43.7	—	71.4
InternVL-2	97.1	49.8	—	78.9
LISA	101.4	52.3	64.1	74.2
PixelLM	107.9	55.0	67.8	76.5
AgriScope	118.6	59.4	73.2	82.7

CIDEr and mIoU for grounded caption generation (GCG). LLaVA and InternVL-2 do not support pixel-level output (Seg F1 not applicable). AgriScope is the first model to achieve over 59 mIoU on this benchmark.

Ablations and Interpretation

Standard ViT encoder (no BSE): CIDEr −6.8; the BSE’s biological specificity is critical for fine-grained disease staging.

Single-resolution features (no FPN): mIoU −3.1; multi-scale features are needed to localise both large leaf lesions and small pest bodies.

Without AgriGround (existing datasets only): CIDEr −14.3; breadth and scale of agricultural supervision are the primary driver.

Dataset scaling: 100K samples → 250K: +6.2 CIDEr; 250K → 500K: +2.1 CIDEr. Diminishing returns but still positive, suggesting continued benefit from data collection.

Decoder variant: Mask transformer vs. FPN upsampling: +2.7 mIoU; mask transformer’s cross-attention to text queries is essential for language-conditioned segmentation.

Reference

Abderrahmene Boudiaf, Mohamad Alanssari, Irfan Husain, Sajid Javed. **AgriScope: Pixel-Grounded Multimodal Understanding for Agricultural Images**. arXiv:2609.20325, September 2026. ICML 2026. <https://arxiv.org/abs/2609.20325>